Zaki Leghtas

Professor, Mines Paris – PSL · Laboratoire de Physique de l'ENS · QUANTIC team, Inria Paris

Personal information

ORCID: 0000-0002-9172-1537
Date of birth: 25 December 1986
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Highlights

- **Publications:** 53 papers (49 peer-reviewed, 4 preprints) · h-index 33 · 10,195 citations (Google Scholar, September 2026)
- **High-impact journals:** 5 *Nature*, 2 *Science*, 2 *Nature Physics*, 5 *Nature Communications*, 8 *Physical Review X*, 2 *PRX Quantum*, 5 *Physical Review Letters*
- **Landmark results on cat qubits:**
  - *Science* 2015 (first author) — confining light to a quantum manifold by engineered two-photon loss, the founding experiment of dissipative cat qubits
  - *Nature Physics* 2020 (last author) — exponential suppression of bit-flips in a qubit encoded in an oscillator
  - *Nature* 2024 (last author) — cat qubit with bit-flip times exceeding ten seconds
- **Technology transfer:** the cat-qubit technology developed in my group is at the origin of Alice & Bob, co-founded in 2020 by my former PhD student Raphaël Lescanne; member of its scientific board since its creation. The company has raised €130M and employs over 180 people.
- **Intellectual property:** 3 patents and patent applications (cat qubits, Josephson circuit elements, microwave photon detection)
- **Funding:** €4.1M in competitive research grants, including an **ERC Starting Grant** (€1.5M, 2020–2025) and BPI iDemo (€1.1M)
- **Recognition:** Top 100 Innovators of the Year, *Le Point* (2026); ~40 invited talks (incl. APS March Meeting ×3, Les Houches, Quantum Cavities); 15 PhD/HDR juries in Europe and Canada
- **Mentoring:** 9 PhD students (7 graduated) and 2 postdocs

Current position

2015–present Professor – Mines Paris

Laboratoire de Physique de l'École Normale Supérieure, Paris
QUANTIC team, INRIA Paris

Previous position

2012–2015 Postdoctoral researcher – Yale University

Group of Michel Devoret
Driven dissipative superconducting quantum circuits (experiments)

Education

2020 Research Direction Habilitation – École Normale Supérieure Paris

2009–2012 PhD – Quantum state engineering and stabilization (theory)

Mines Paris / INRIA Paris
Supervisors: Pierre Rouchon and Mazyar Mirrahimi

2006–2009 Master of Science and Engineering – Mines Paris

Teaching activities

- 2026–present — Quantum Technologies – Mines Paris PSL Executive Education (32h/year)
- 2025–present — Quantum Information – EMINES UM6P Morocco (30h/year)
- 2023–present — Circuit QED – M2 ICFP (18h/year)
- 2018–2022 — Quantum Information – Mines Paris
- 2016–2021 — Quantum and Statistical Physics – Mines Paris
- 2016–2018 — Quantum Physics – PSL

Grants

- 2024–2027 — BPI iDemo — €1.1M
- 2022–2027 — PEPR Plan Quantique — €500k
- 2020–2025 — ERC Starting Grant — €1.5M
- 2020–2023 — Quantera — €300k
- 2018 — Sirteq — €50k
- 2016–2019 — Emergence Ville de Paris — €230k
- 2016–2020 — ANR Accueil Chercheur Haut Niveau — €400k

Awards

- Top 100 Innovators of the Year – Le Point (2026)

Supervision

Students

- Armelle Célarier
- Erwan Roverc'h

Postdocs

- Kyrylo Gerashchenko

Alumni

- Alvise Borgognoni
- Adrien Bocquet
- Ulysse Réglade
- Marius Villiers
- Clarke Smith
- Camille Berdou
- Raphaël Lescanne
- Lucas Verney

Scientific boards and committees

- Scientific board – Alice & Bob
- Scientific committee – Quantum Information Center, Sorbonne doctoral school

Organization of scientific meetings

- Nov 2018 — Organizing committee, "International Conference on Quantum Computation" (ICoQC 2018), École Normale Supérieure, Paris, France

PhD defense committees

- 2026 — Guillaume Beaulieu, EPFL (invited by Pasquale Scarlino)
- 2026 — Alexandru Petrescu, ENS Paris (HDR)
- 2026 — Joan Cáceres, CEA Saclay (invited by Hugues Pothier)
- 2026 — Desislava G. Atanasova, Innsbruck University (invited by Gerhard Kirchmair)
- 2026 — Simon Bolduc Beaudoin, Université de Sherbrooke (invited by Bertrand Reulet)
- 2025 — Alexander Wagner, CEA Saclay (invited by Caglar Girit)
- 2025 — Jacob Koenig, TU Delft (invited by Gary Steele)
- 2023 — Jean-Damien Pillet, École Polytechnique Paris (HDR)
- 2022 — Catherine Leroux, Université de Sherbrooke (invited by Alexandre Blais)
- 2022 — Guilliam Butseraen, Institut Néel Grenoble (invited by Julien Renard)
- 2021 — Daniel Garcia-Sanchez, INSP Paris (HDR)
- 2021 — Sébastien Leger, Institut Néel Grenoble (invited by Nicolas Roch)
- 2020 — Mario Gely, TU Delft (invited by Gary Steele)
- 2019 — Romain Albert, CEA Grenoble (invited by Max Hofheinz)
- 2018 — Iivari Pietikäinen, University of Oulu (invited by Erkki Thuneberg)

Invited seminars

- 2026 — Colloquium, École Normale Supérieure, Paris (invited by Christina Psaroudaki
- 2026 — DigiQ, École Normale Supérieure, Paris (invited by Tarik Yefsah and Clément Sayrin)
- 2025 — Walther Meissner Institute, Garching, Germany (invited by Stefan Philipp)
- 2024 — Sorbonne Université (invited by Nabil Guarroum)
- 2024 — Université Paris Diderot (invited by Edouard Boulat)
- 2024 — Institute of Science and Technology Austria (invited by Johannes Fink)
- 2023 — Hebrew University of Jerusalem (invited by Alex Retzker)
- 2023 — Weizmann Institute (invited by Fabien Lafont and Serge Rosenblum)
- 2023 — Institut Néel (invited by Jérémie Viennot)
- 2023 — C2N (invited by Rebeca Ribeiro-Palau)
- 2022 — Sorbonne Université (invited by Julien Laurat)
- 2022 — LJLL (invited by Emmanuel Trélat)
- 2022 — Journées X/ENS (invited by Frédéric Chevy)
- 2022 — ENS (invited by Carlo Sirtori)
- 2022 — ENSTA (invited by Davide Boschetto)
- 2021 — IBM Qiskit, online (invited by Zlatko Minev)
- 2020 — École Polytechnique, France (invited by Timothée Nicolas)
- 2018 — University of Oulu, Finland (invited by Erkki Thuneberg)
- 2018 — University of Berkeley, USA (invited by Irfan Siddiqi)
- 2018 — Rigetti Quantum Computing, Berkeley, USA (invited by Chad Rigetti)
- 2016 — KIT, Karlsruhe, Germany (invited by Ioan Pop)
- 2016 — Néel Institute, Grenoble, France (invited by Nicolas Roch)
- 2016 — SPEC, CEA Saclay, France (invited by Daniel Esteve)

Invited conferences

- 2026 — Dynamics and control of classical and quantum systems, Mines Paris (invited by Pierre Rouchon)
- 2026 — Quantum Cavities, Gothenburg, Sweden (invited by Per Delsing and Goran Johansson)
- 2026 — Quantum Matter, Barcelona, Spain (invited by Stephan Roche)
- 2024 — Frontiers of Condensed Matter, Les Houches, France (invited by Julia Meyer)

- 2024 — Quantum Cavities, Canada (invited by Alexandre Blais and Aashish Clerk)
- 2023 — Les Houches, France (invited by Benjamin Sacépé)
- 2022 — Quantum Innovation, Tokyo, online (invited by Yutaka Tabuchi)
- 2022 — APS March Meeting, Chicago, online (invited by Arne Grimsmo)
- 2021 — Cargèse Méso School, Corsica (invited by Benjamin Huard)
- 2020 — Neuromorphic meets quantum, C2N Palaiseau, France (invited by Danijela Markovic)
- 2020 — Byron Bay Quantum Workshop, online (invited by Arne Grimsmo)
- 2019 — GDR mésoscopique, Aussois, France (invited by Bernard Plaçais)
- 2019 — Quantum Cavities, Jouvence, Canada (invited by Alexandre Blais)
- 2019 — APS March Meeting, Boston, USA (invited by Blake Johnson)
- 2018 — ONE-QOS Workshop, Max Planck, Erlangen, Germany (invited by Florian Marquardt)
- 2018 — IHP Workshop on quantum control and feedback, Paris, France (invited by Eleni Diamanti)
- 2018 — LIA CNRS–Université de Sherbrooke workshop, Saint-Rémy, France (invited by Denis Vion)
- 2015 — APS March Meeting, San Antonio, Texas, USA (invited by Eliott Kapit)

Peer review

- Journal articles: CR de l'Académie des Sciences, PRL, PRX, Nature Communications, Nature Physics, Nature
- Grants: ERC Starting Grants

Intellectual property

- Asymmetric repetition code for a cat qubit — Patent US20250190839A1
- Josephson dipole element mediating two-to-one photon exchange — Patent US11302856B2
- Microwave photon detector — Patent US20220057261A1

Publications

2026

[53] E. Roverc'h, A. Borgognoni, M. Villiers, K. Gerashchenko, W. C. Smith, C. Wilson, B. Douçot, A. Petrescu, P. Campagne-Ibarcq, Z. Leghtas. *Experimental realization of a cos(2φ) transmon qubit.* Preprint (2026). [arXiv:2603.13114]

[52] A. Vanselow, B. Beauseigneur, L. Lattier, M. Villiers, A. Denis, P. Morfin, Z. Leghtas, P. Campagne-Ibarcq. *Dissipating quartets of excitations in a superconducting circuit.* Phys. Rev. X 16, 011032 (2026). [arXiv:2501.05960]

2025

[51] M. Praquin, A. Giraudo, V. Lienhard, T. Bouwakdh, A. Vanselow, Z. Leghtas, P. Campagne-Ibarcq. *Mixing of counterpropagating signals in a traveling-wave Josephson device.* Nature Communications 16, 11390 (2025). [doi:10.1038/s41467-025-66190-0] [arXiv:2406.19751]

[50] R. Dhundhwal, H. Duan, L. Brauch, S. Arabi, D. Fuchs, A.-A. Haghighirad, A. Welle, F. Scharwaechter, S. Pal, M. Scheffler, J. Palomo, Z. Leghtas, A. Murani, H. Hahn, J. Aghassi-Hagmann, C. Kübel, W. Wulfhekel, I. M. Pop, T. Reisinger. *High-quality superconducting tantalum resonators with beta phase defects.* Applied Physics Letters 127, 214005 (2025). [arXiv:2502.17247]

[49] R. Rousseau, D. Ruiz, E. Albertinale, P. d'Avezac, D. Banys, U. Blandin, N. Bourdaud, G. Campanaro, G. Cardoso, N. Cottet, C. Cullip, S. Deléglise, L. Devanz, A. Devulder, A. Essig, P. Février, A. Gicquel, É. Gouzien, A. Gras, J. Guillaud, E. Gümüş, M. Hallén, A. Jacob, P. Magnard, A. Marquet, S. Miklass, T. Peronnin, S. Polis, F. Rautschke, U. Réglade, J. Roul, J. Stevens, J. Solard, A. Thomas, J.-L. Ville, P. Wan-Fat, R. Lescanne, Z. Leghtas, J. Cohen, S. Jezouin, A. Murani. *Enhancing dissipative cat qubit protection by squeezing.* Preprint (2025). [arXiv:2502.07892]

[48] L.-A. Sellem, A. Sarlette, Z. Leghtas, M. Mirrahimi, P. Rouchon, P. Campagne-Ibarcq. *Dissipative protection of a GKP qubit in a high-impedance superconducting circuit driven by a microwave frequency comb.* Physical Review X 15, 011011 (2025). [arXiv:2304.01425]

[47] W. C. Smith, A. Borgognoni, M. Villiers, E. Roverc'h, J. Palomo, M. R. Delbecq, T. Kontos, P. Campagne-Ibarcq, B. Doucot, Z. Leghtas. *Spectral signature of high-order photon processes enhanced by Cooper-pair pairing.* Nature Communications 16, 8359 (2025). [arXiv:2312.15075]

[46] K. Gerashchenko, R. Rousseau, L. Balembois, H. Patange, P. Manset, T. Briant, P.-F. Cohadon, A. Heidmann, W. C. Smith, A. Tilloy, Z. Leghtas, E. Flurin, T. Jacqmin, S. Deléglise. *Probing the quantum motion of a macroscopic mechanical oscillator with a radio-frequency superconducting qubit.* Preprint (2025). [arXiv:2505.21481]

2024

[45] U. Reglade, A. Bocquet, R. Gautier, J. Cohen, A. Marquet, E. Albertinale, N. Pankratova, M. Hallen, F. Rautschke, L.-A. Sellem, P. Rouchon, A. Sarlette, M. Mirrahimi, P. Campagne-Ibarcq, R. Lescanne, S. Jezouin, Z. Leghtas. *Quantum control of a cat-qubit with bit-flip times exceeding ten seconds.* Nature 629, 778–783 (2024). [arXiv:2307.06617]

[44] D. Willsch et al. *Observation of Josephson Harmonics in Tunnel Junctions.* Nature Physics 20, 815 (2024). [arXiv:2302.09192]

[43] B.-L. Najera-Santos, R. Rousseau, K. Gerashchenko, H. Patange, A. Riva, M. Villiers, T. Briant, P.-F. Cohadon, A. Heidmann, J. Palomo, M. Rosticher, H. le Sueur, A. Sarlette, W. C. Smith, Z. Leghtas, E. Flurin, T. Jacqmin, S. Deleglise. *High-sensitivity AC-charge detection with a MHz-frequency fluxonium qubit.* Phys. Rev. X 14, 011007 (2024).

[42] M. Villiers, W. C. Smith, A. Petrescu, A. Borgognoni, M. Delbecq, A. Sarlette, M. Mirrahimi, P. Campagne-Ibarcq, T. Kontos, Z. Leghtas. *Dynamically enhancing qubit-photon interactions with anti-squeezing.* PRX Quantum 5, 020306 (2024). [arXiv:2212.04991]

2023

[41] C. Berdou, A. Murani, U. Reglade, W. C. Smith, M. Villiers, J. Palomo, M. Rosticher, A. Denis, P. Morfin, M. Delbecq, T. Kontos, N. Pankratova, F. Rautschke, T. Peronnin, L.-A. Sellem, P. Rouchon, A. Sarlette, M. Mirrahimi, P. Campagne-Ibarcq, S. Jezouin, R. Lescanne, Z. Leghtas. *One hundred second bit-flip time in a two-photon dissipative oscillator.* PRX Quantum 4, 020350 (2023). [arXiv:2204.09128]

[40] F.-M. Le Regent, C. Berdou, Z. Leghtas, J. Guillaud, M. Mirrahimi. *High-performance repetition cat code using fast noisy operations.* Quantum 7, 1198 (2023). [arXiv:2212.11927]

[39] L. C. Contamin, T. Cubaynes, W. Legrand, M. Margańska, M. Villiers, M. R. Delbecq, A. Cottet, Z. Leghtas, T. Kontos. *Inhomogeneous magnetic fields interacting with spinful states in a double quantum dot: Evidence for a staggered spin-orbit interaction.* Physical Review B 107, 085152 (2023).

2022

[38] W. C. Smith, M. Villiers, A. Marquet, J. Palomo, M. R. Delbecq, T. Kontos, P. Campagne-Ibarcq, B. Doucot, Z. Leghtas. *Magnifying quantum phase fluctuations with Cooper-pair pairing.* Phys. Rev. X 12, 021002 (2022). [arXiv:2010.15488]

2021

[37] Z. K. Minev, Z. Leghtas, S. O. Mundhada, L. Christakis, I. M. Pop, M. H. Devoret. *Energy-participation quantization of Josephson circuits.* npj Quantum Information 7, 131 (2021). [arXiv:2010.00620]

[36] A. Essig, Q. Ficheux, T. Peronnin, N. Cottet, R. Lescanne, A. Sarlette, P. Rouchon, Z. Leghtas, B. Huard. *Multiplexed photon number measurement.* Physical Review X 11, 031045 (2021). [arXiv:2001.03217]

[35] F. Valmorra, K. Yoshida, L. C. Contamin, S. Messelot, S. Massabeau, M. R. Delbecq, M. C. Dartiailh, M. M. Desjardins, T. Cubaynes, Z. Leghtas, K. Hirakawa, J. Tignon, S. Dhillon, S. Balibar, J. Mangeney, A. Cottet, T. Kontos. *Vacuum-field-induced THz transport gap in a carbon nanotube quantum dot.* Nature Communications 12 (2021).

2020

[34] R. Lescanne, M. Villiers, T. Peronnin, A. Sarlette, M. Delbecq, B. Huard, T. Kontos, M. Mirrahimi, Z. Leghtas. *Exponential suppression of bit-flips in a qubit encoded in an oscillator.* Nature Physics 16, 509–513 (2020). [arXiv:1907.11729]

[33] R. Lescanne, S. Deleglise, E. Albertinale, U. Reglade, T. Capelle, E. Ivanov, T. Jacqmin, Z. Leghtas, E. Flurin. *Irreversible Qubit-Photon Coupling for the Detection of Itinerant Microwave Photons.* Phys. Rev. X 10, 021038 (2020). [arXiv:1902.05102]

[32] A. Cottet, Z. Leghtas, T. Kontos. *Theory of interactions between cavity photons induced by a mesoscopic circuit.* Phys. Rev. B 102, 155105 (2020). [arXiv:1910.07801]

2019

[31] T. Cubaynes, M. R. Delbecq, M. C. Dartiailh, R. Assouly, M. M. Desjardins, L. C. Contamin, L. E. Bruhat, Z. Leghtas, F. Mallet, A. Cottet, T. Kontos. *Highly coherent spin states in carbon nanotubes coupled to cavity photons.* NPJ Quantum Information 5, 47 (2019). [arXiv:1903.05229]

[30] L. Verney, R. Lescanne, M. H. Devoret, Z. Leghtas, M. Mirrahimi. *Structural instability of driven Josephson circuits prevented by an inductive shunt.* Phys. Rev. Applied 11, 024003 (2019). [arXiv:1805.07542]

[29] R. Lescanne, L. Verney, Q. Ficheux, M. H. Devoret, B. Huard, M. Mirrahimi, Z. Leghtas. *Observing the escape of a driven quantum Josephson circuit into unconfined states.* Phys. Rev. Applied 11, 014030 (2019). [arXiv:1805.05198]

2018

[28] S. Touzard, A. Grimm, Z. Leghtas, S. O. Mundhada, P. Reinhold, R. Heeres, C. Axline, M. Reagor, K. Chou, J. Blumoff, K. M. Sliwa, S. Shankar, L. Frunzio, R. J. Schoelkopf, M. Mirrahimi, M. H. Devoret. *Coherent oscillations inside a quantum manifold stabilized by dissipation.* Phys. Rev. X 8, 021005 (2018). [arXiv:1705.02401]

[27] Q. Ficheux, S. Jezouin, Z. Leghtas, B. Huard. *Dynamics of a qubit while simultaneously monitoring its relaxation and dephasing.* Nature Communications 9, 1926 (2018). [arXiv:1711.01208]

[26] D. Markovic, S. Jezouin, Q. Ficheux, S. Fedortchenko, S. Felicetti, T. Coudreau, P. Milman, Z. Leghtas, B. Huard. *Demonstration of an effective ultrastrong coupling between two oscillators.* Phys. Rev. Lett. 121, 040505 (2018). [arXiv:1804.08705]

[25] E. Zalys-Geller, A. Narla, S. Shankar, M. Hatridge, M. P. Silveri, K. Sliwa, Z. Leghtas, M. H. Devoret. *Generation of discord through a remote joint continuous variable measurement.* Preprint (2018). [arXiv:1803.01275]

2016

[24] N. Ofek, A. Petrenko, R. Heeres, P. Reinhold, Z. Leghtas, B. Vlastakis, Y. Liu, L. Frunzio, S. M. Girvin, L. Jiang, M. Mirrahimi, M. H. Devoret, R. J. Schoelkopf. *Extending the lifetime of a quantum bit with error correction in superconducting circuits.* Nature 536, 441–445 (2016). [arXiv:1602.04768]

[23] A. Narla, S. Shankar, M. Hatridge, Z. Leghtas, K. M. Sliwa, E. Zalys-Geller, S. O. Mundhada, W. Pfaff, L. Frunzio, R. J. Schoelkopf, M. H. Devoret. *Robust concurrent remote entanglement between two superconducting qubits.* Phys. Rev. X 6, 031036 (2016). [arXiv:1603.03742]

[22] M. Silveri, E. Zalys-Geller, M. Hatridge, Z. Leghtas, M. H. Devoret, S. M. Girvin. *Theory of remote entanglement via quantum-limited phase-preserving amplification.* Phys. Rev. A 93, 062310 (2016). [arXiv:1507.00732]

[21] Z. K. Minev, K. Serniak, I. M. Pop, Z. Leghtas, K. Sliwa, M. Hatridge, L. Frunzio, R. J. Schoelkopf, M. H. Devoret. *Planar multilayer circuit quantum electrodynamics.* Phys. Rev. Applied 5, 044021 (2016). [arXiv:1509.01619]

2015

[20] Z. Leghtas, S. Touzard, I. M. Pop, A. Kou, B. Vlastakis, A. Petrenko, K. M. Sliwa, A. Narla, S. Shankar, M. J. Hatridge, M. Reagor, L. Frunzio, R. J. Schoelkopf, M. Mirrahimi, M. H. Devoret. *Confining the state of light to a quantum manifold by engineered two-photon loss.* Science 347, 853–857 (2015). [arXiv:1412.4633]

[19] B. Vlastakis, A. Petrenko, N. Ofek, L. Sun, Z. Leghtas, K. Sliwa, Y. Liu, M. Hatridge, J. Blumoff, L. Frunzio, M. Mirrahimi, L. Jiang, M. H. Devoret, R. J. Schoelkopf. *Characterizing entanglement of an artificial atom and a cavity cat state with Bell's inequality.* Nature Communications 6, 8970 (2015). [arXiv:1504.02512]

[18] E. T. Holland, B. Vlastakis, R. W. Heeres, M. J. Reagor, U. Vool, Z. Leghtas, L. Frunzio, G. Kirchmair, M. H. Devoret, M. Mirrahimi, R. J. Schoelkopf. *Single-Photon-Resolved Cross-Kerr Interaction for Autonomous Stabilization of Photon-Number States.* Phys. Rev. Lett. 115, 180501 (2015). [arXiv:1504.03382]

[17] A. Roy, Z. Leghtas, A. D. Stone, M. Devoret, M. Mirrahimi. *Continuous generation and stabilization of mesoscopic field superposition states in a quantum circuit.* Physical Review A 91, 013810 (2015).

2014

[16] L. Sun, A. Petrenko, Z. Leghtas, B. Vlastakis, G. Kirchmair, K. M. Sliwa, A. Narla, M. Hatridge, S. Shankar, J. Blumoff, L. Frunzio, M. Mirrahimi, M. H. Devoret, R. J. Schoelkopf. *Tracking photon jumps with repeated quantum non-demolition parity measurements.* Nature 511, 444 (2014). [arXiv:1311.2534]

[15] M. Mirrahimi, Z. Leghtas, V. V. Albert, S. Touzard, R. J. Schoelkopf, L. Jiang, M. H. Devoret. *Dynamically protected cat-qubits: a new paradigm for universal quantum computation.* New Journal of Physics 16, 045014 (2014). [arXiv:1312.2017]

2013

[14] G. Kirchmair, B. Vlastakis, Z. Leghtas, S. E. Nigg, H. Paik, E. Ginossar, M. Mirrahimi, L. Frunzio, S. M. Girvin, R. J. Schoelkopf. *Observation of quantum state collapse and revival due to the single-photon Kerr effect.* Nature 495, 205–209 (2013). [arXiv:1211.2228]

[13] S. Shankar, M. Hatridge, Z. Leghtas, K. M. Sliwa, A. Narla, U. Vool, S. M. Girvin, L. Frunzio, M. Mirrahimi, M. H. Devoret. *Autonomously stabilized entanglement between two superconducting quantum bits.* Nature 504 (2013). [arXiv:1307.4349]

[12] B. Vlastakis, G. Kirchmair, Z. Leghtas, S. E. Nigg, L. Frunzio, S. M. Girvin, M. Mirrahimi, M. H. Devoret, R. J. Schoelkopf. *Deterministically encoding quantum information using 100-photon Schrödinger cat states.* Science 342, 607–610 (2013).

[11] Z. Leghtas, G. Kirchmair, B. Vlastakis, R. J. Schoelkopf, M. H. Devoret, M. Mirrahimi. *Hardware-efficient autonomous quantum memory protection.* Physical Review Letters 111, 120501 (2013).

[10] Z. Leghtas, U. Vool, S. Shankar, M. Hatridge, S. M. Girvin, M. H. Devoret, M. Mirrahimi. *Stabilizing a Bell state of two superconducting qubits by dissipation engineering.* Physical Review A 88, 023849 (2013).

[9] Z. Leghtas, G. Kirchmair, B. Vlastakis, M. H. Devoret, R. J. Schoelkopf, M. Mirrahimi. *Deterministic protocol for mapping a qubit to coherent state superpositions in a cavity.* Physical Review A 87, 042315 (2013).

[8] K. Geerlings, Z. Leghtas, I. M. Pop, S. Shankar, L. Frunzio, R. J. Schoelkopf, M. Mirrahimi, M. H. Devoret. *Demonstrating a driven reset protocol for a superconducting qubit.* Physical Review Letters 110, 120501 (2013). [arXiv:1211.0491]

[7] R. Rey-de Castro, Z. Leghtas, H. Rabitz. *Manipulating quantum pathways on the fly.* Physical Review Letters 110, 223601 (2013).

2012

[6] A. Sarlette, Z. Leghtas, M. Brune, J. M. Raimond, P. Rouchon. *Stabilization of nonclassical states of one- and two-mode radiation fields by reservoir engineering.* Physical Review A 86, 012114 (2012). [arXiv:1203.0929]

[5] Z. Leghtas, G. Turinici, H. Rabitz, P. Rouchon. *Hamiltonian identification through enhanced observability utilizing quantum control.* IEEE Transactions on Automatic Control 57, 2679–2683 (2012).

[4] O. M. Shir, J. Roslund, Z. Leghtas, H. Rabitz. *Quantum control experiments as a testbed for evolutionary multi-objective algorithms.* Genetic Programming and Evolvable Machines 13, 445–491 (2012).

2011

[3] Z. Leghtas, A. Sarlette, P. Rouchon. *Adiabatic passage and ensemble control of quantum systems.* Journal of Physics B 44, 154017 (2011).

[2] Z. Leghtas, M. Mirrahimi, P. Rouchon. *Back and forth nudging for quantum state estimation by continuous weak measurement.* American Control Conference (2011).

2009

[1] Z. Leghtas, M. Mirrahimi, P. Rouchon. *Parameter estimation of a 3-level quantum system with a single population measurement.* IEEE Conference on Decision and Control (2009).